

	Odd semester <u>Examination</u> – Academic year 2023/2024			
	Subject	Physics II		
	Class	B₁CE-B₁CSE	Date	11/ 06 /2024
	Max. Marks	50	Time	2 hours
	Lecturer	Dr. Othman – Eng. Ala'a	Signature	
Student Name		Class		Reg. No

$g = 10 \text{ m/s}^2$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$	$K_e = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$, $G = 6.67 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$
Proton: $q = 1.6 \times 10^{-19} \text{ C}$, $m = 1.67 \times 10^{-27} \text{ kg}$	Electron: $q = - 1.6 \times 10^{-19} \text{ C}$, $m = 9.11 \times 10^{-31} \text{ kg}$

Section A: Answer the following questions with **True (T)** or **False (F)**: **14 Marks**

1- A current flows into a paper, a direction of B is clockwise	(T)	()
2- The magnetic flux through closed surface is q_{in}/ϵ_0	()	(F)
3- The electric field can change the magnitude of the velocity	(T)	()
4- If a charge enters parallel to B, the magnetic field is zero	()	(F)
5- The electron has negative charge	(T)	()
6- If a mass hanged to a spring increases, the period increases	(T)	()
7- The magnetic field is minimum at the center	(T)	()

Section B: Select the proper choice for each of the following: **16 Marks**

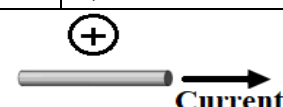
8- In SHM: The period can be given as:

a) $2\pi\sqrt{\frac{k}{m}}$	b) $2\pi\sqrt{\frac{m}{k}}$	c) $2\pi/\sqrt{\frac{m}{k}}$	d) $\sqrt{\frac{k}{m}}/2\pi$	e) None of these
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9- Which of the following represents a magnetic force:

a) $q \vec{v} \times \vec{B}$	b) $I B L \sin \theta$	c) $I B v \sin \theta$	d) $I \vec{v} \times \vec{B}$	e) None of these
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10- A magnetic force direction on a charge moving left as in Figure is:



a) + x	b) + z	c) + y	d) - z	e) None of these
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11- To make a neutral object having negative charge, we add

a) Photons	b) Electrons	c) Neutrons	d) Protons	e) None of these
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12- If $x(t) = (4 \text{ m}) \cos (4\pi t + 7\pi)$. The angular velocity is

a) 7π	b) $7\pi t$	c) 4π	d) $4\pi t$	e) None of these
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13- We represent the electric field by:

a) F	b) q	c) B	d) E	e) None of these
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14- A velocity of charge increases when a charge enters a field to B

a) Parallel	b) $0^\circ < \theta < 90^\circ$	c) Anti parallel	d) Perpendicular	e) None of these
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15- Two charges on x axis: q_1 at $x = a$ and q_2 at $x = - 2a$. The net force on a third charge placed at the $x = 0$ is zero, q_1 and q_2 must be related by:

a) $q_2 = 4 q_1$	b) $q_2 = - 2q_1$	c) $q_2 = q_1/4$	d) $q_2 = - q_1/2$	e) None of these
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Section C: Solve the following problems and select the proper choice:

20 Marks

Note: Show the full steps clearly for problems 19 and 20

16- In SHM: If a period is 0.02 s. The frequency is:

a) 0.05 Hz	b) 5 Hz	c) 50 Hz	d) 0.5 Hz	e) None of these
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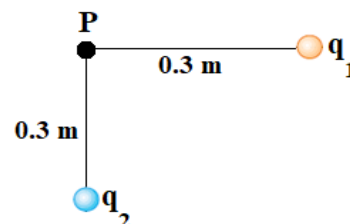
17- A charge of - 4 C moves with velocity of 20 m/s directed to z axis in magnetic field of 3 T directed to x axis. The magnitude and direction of magnetic force is:

a) -24 N at y	b) 240 N at y	c) 24 N at -y	d) 240 N at -y	e) None of these
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18- A block whose mass $m = 250$ g is fastened to a spring with k is 8 N/m.

a) 8 rad/s	b) 3 rad/s	c) 2 rad/s	d) 4 rad/s	e) None of these
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19- Given that $q_1 = q_2 = 5$ nC in the figure. The magnitude of the total electric field at P is:

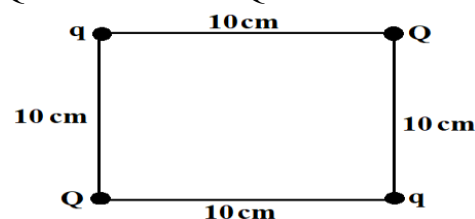


a) 565.7 N/C	b) 707.7 N/C	c) 424.3 N/C	d) 500000 N/C	e) None of these
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20- 9 electrons and 2 protons have a total charge of:

a) 11.2×10^{-19} C	b) -9.6×10^{-19} C	c) 9.6×10^{-19} C	d) -11.2×10^{-19} C	e) None of these
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21- A square is shown. If the total electric force on a charge Q is zero. Then Q is:



a) 2.83 q	b) $-q/4.33$	c) $-5.66 q/2$	d) $6.33/q$	e) None of these
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22- A wire has current of 24 A and 0.75 m along the y within a magnetic field $\vec{B} = 1.6 \hat{k}$ (T). The magnetic force on the wire is:

a) $-28.8 \hat{i}$ (N)	b) $-28.8 \hat{j}$ (N)	c) $28.8 \hat{i}$ (N)	d) $28.8 \hat{j}$ (N)	e) None of these
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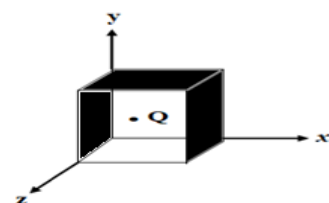
23- A wire with mass per unit length of 0.04 kg/cm carries a current of 2 A moving to the right placed in magnetic field of 2 T directed outward a paper. The tension in the rope required to make the wire stable above the ground is 30.8 N. The length of wire will be:

a) 0.7 m	b) 7.7 m	c) 70 m	d) 7 m	e) None of these
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24- A proton moving with a velocity of $\vec{v} = 4 \hat{i} + 8 \hat{j} + 2 \hat{k}$ (m/s), in $\vec{B} = 2 \hat{i} + 4 \hat{j} - 6 \hat{k}$ (T). The magnitude of the magnetic force is:

a) 31.9×10^{-19} N	b) 71.6×10^{-19} N	c) 69.2×10^{-19} N	d) 100.2×10^{-19} N	e) None of these
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25- If $Q = 53.1 \times 10^{-11}$ C. The electric flux of the 3 shaded faces of the cube is:



a) 30 N.m ² /C	b) 2 N.m ² /C	c) 3 N.m ² /C	d) 3 N.m ² /C	e) None of these
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	Odd semester Examination – Academic year 2023/2024			
	Subject	Physics II		
	Class	B₁ME-B₁MIE	Date	11/ 06 /2024
	Max. Marks	50	Time	2 hours
	Lecturer	Dr. Mehdi Dabban – Eng. Ala'a	Signature	
Student Name		Class		Reg. No

$g = 10 \text{ m/s}^2$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$	$K_e = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$, $G = 6.67 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$
Proton: $q = 1.6 \times 10^{-19} \text{ C}$, $m = 1.67 \times 10^{-27} \text{ kg}$	Electron: $q = - 1.6 \times 10^{-19} \text{ C}$, $m = 9.11 \times 10^{-31} \text{ kg}$

Section A: Answer the following questions with **True (T)** or **False (F)**: **14 Marks**

1- A current flows out of a paper, a direction of B is counter clockwise	(T)	()
2- The proton has negative charge	()	(F)
3- The electric flux through closed surface is q_{in}/ϵ_0	(T)	()
4- Magnetic field is minimum at the poles	()	(F)
5- If a charge enters parallel to B, the magnetic field is zero	()	(F)
6- The magnitude of the velocity can be changed by electric force	(T)	()
7- The frequency decreases as the mass hanged to the spring increases	(T)	()

Section B: Select the proper choice for each of the following: **16 Marks**

8- A velocity of charge increases when a charge enters a field to B

a) Perpendicular	b) Anti parallel	c) $0^\circ < \theta < 90^\circ$	d) Parallel	e) None of these
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9- If $x(t) = (4 \text{ m}) \cos (5\pi t + 3\pi)$. The angular velocity is

a) 3π	b) 5π	c) $5\pi t$	d) 3π	e) None of these
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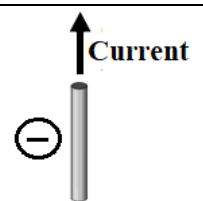
10- We represent the force by:

a) F	b) q	c) B	d) E	e) None of these
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11- To make a neutral object having positive charge, we remove

a) Protons	b) Photons	c) Neutrons	d) Electrons	e) None of these
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12- A magnetic force direction on charge moving up as in Figure is:



a) + y	b) - x	c) + x	d) - z	e) None of these
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13- In SHM: The period can be given as:

a) $\sqrt{\frac{k}{m}}/2\pi$	b) $2\pi\sqrt{\frac{k}{m}}$	c) $2\pi\sqrt{\frac{m}{k}}$	d) $2\pi/\sqrt{\frac{m}{k}}$	e) None of these
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14- Which of the following represents a magnetic force:

a) $q \vec{v} \times \vec{E}$	b) $q \vec{B} \times \vec{v}$	c) $q B v \sin \theta$	d) $q \vec{L} \times \vec{B}$	e) None of these
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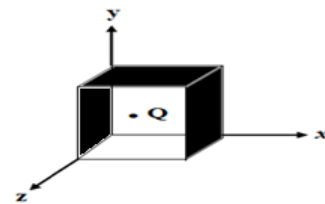
15- Two charges on x axis: q_1 at $x = a$ and q_2 at $x = - 2a$. The net force on a third charge placed at the $x = 0$ is zero, q_1 and q_2 must be related by:

a) $q_2 = q_1/2$	b) $q_1 = q_2/4$	c) $q_1 = 4 q_2$	d) $q_1 = - 2q_2$	e) None of these
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Section C: Solve the following problems and select the proper choice:

20 Marks

16- If $Q = 53.1 \times 10^{-10}$ C. The electric flux of the 3 shaded faces of the cube is:



a) $30 \text{ N.m}^2/\text{C}$	b) $2 \text{ N.m}^2/\text{C}$	c) $3 \text{ N.m}^2/\text{C}$	d) $3 \text{ N.m}^2/\text{C}$	e) None of these
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17- A proton moving with a velocity of $\vec{v} = 4\hat{i} + 8\hat{j} + 2\hat{k}$ (m/s) in $\vec{B} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ (T). The magnitude of the magnetic force is:

a) $31.9 \times 10^{-19} \text{ N}$	b) $71.6 \times 10^{-19} \text{ N}$	c) $69.2 \times 10^{-19} \text{ N}$	d) $100.2 \times 10^{-19} \text{ N}$	e) None of these
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18- A block whose mass $m = 500$ g is fastened to a spring with k is 2 N/m . The angular velocity is:

a) 8 rad/s	b) 3 rad/s	c) 2 rad/s	d) 4 rad/s	e) None of these
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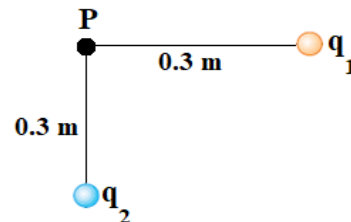
19- A wire with mass per unit length of 0.04 kg/cm carries a current of 2 A moving to the right placed in magnetic field of 2 T directed outward a paper. The tension in the rope required to make the wire stable above the ground is 26.4 N . The length of wire will be:

a) 6 m	b) 0.6 m	c) 60 m	d) 6.6 m	e) None of these
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20- A charge of -4 C moves with velocity of 2 m/s directed to z axis in magnetic field of 3 T directed to x axis. The magnitude and direction of magnetic force is:

a) 24 N at $-y$	b) 240 N at y	c) -24 N at y	d) 240 N at $-y$	e) None of these
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21- Given that $q_1 = q_2 = 2 \text{ nC}$ in the figure. The magnitude of the total electric field at P is:



a) 565.7 N/C	b) 707.7 N/C	c) 80000 N/C	d) 282.8 N/C	e) None of these
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22- A wire has current of 24 A and 0.75 m along the $-z$ within a magnetic field $\vec{B} = 1.6\hat{k}$ (T). The magnetic force on the wire is:

a) $-28.8\hat{i} \text{ (N)}$	b) $-28.8\hat{j} \text{ (N)}$	c) $28.8\hat{i} \text{ (N)}$	d) $28.8\hat{j} \text{ (N)}$	e) None of these
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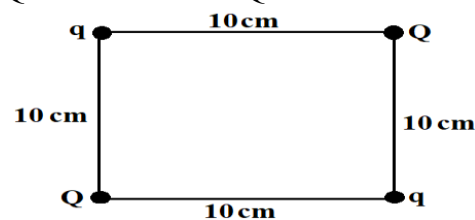
23- 3 electrons and 9 protons have a total charge of:

a) $11.2 \times 10^{-19} \text{ C}$	b) $-9.6 \times 10^{-19} \text{ C}$	c) $9.6 \times 10^{-19} \text{ C}$	d) $-11.2 \times 10^{-19} \text{ C}$	e) None of these
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24- In SHM: If a period is 20 s . The frequency is:

a) 0.05 Hz	b) 5 Hz	c) 50 Hz	d) 0.5 Hz	e) None of these
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25- A square is shown. If the total electric force on a charge Q is zero. Then Q is:



a) $-5.66 q/2$	b) $6.33/q$	c) $2.83 q$	d) $-q/4.33$	e) None of these
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Odd semester Examination – Academic year 2023/2024

Subject Physics II

Class B₁MAR- B₁ECE- B₂MIE

Date 11/ 06 /2024

Max. Marks 50

Time 2 hours

Lecturer Eng. Ala'a

Signature

Student Name

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Class

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Reg. No

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$g = 10 \text{ m/s}^2$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$

$K_e = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$, $G = 6.67 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$

Proton: $q = 1.6 \times 10^{-19} \text{ C}$, $m = 1.67 \times 10^{-27} \text{ kg}$

Electron: $q = - 1.6 \times 10^{-19} \text{ C}$, $m = 9.11 \times 10^{-31} \text{ kg}$

Section A: Answer the following questions with **True (T)** or **False (F)**:

14 Marks

1- A current flows into a paper, a direction of B is counter clockwise	()	(F)
2- The magnetic field is maximum at the center	()	(F)
3- The frequency increases as the mass hanged to the spring increases	()	(F)
4- The electric field can change the magnitude of the velocity	(T)	()
5- If a charge enters parallel to B, the magnetic field is zero	()	(F)
6- The proton has positive charge	(T)	()
7- The magnetic flux through closed surface is zero	(T)	()

Section B: Select the proper choice for each of the following:

16 Marks

8- A velocity of charge increases when a charge enters a field to B

a) Perpendicular	b) Anti parallel	c) $0^\circ < \theta < 90^\circ$	d) Parallel	e) None of these
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9- In SHM: The period can be given as:

a) $2\pi\sqrt{\frac{m}{k}}$	b) $2\pi\sqrt{\frac{k}{m}}$	c) $2\pi/\sqrt{\frac{m}{k}}$	d) $\sqrt{\frac{k}{m}}/2\pi$	e) None of these
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10- We represent the charge by:

a) F	b) q	c) B	d) E	e) None of these
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11- To make a neutral object having negative charge, we remove

a) Protons	b) Neutrons	c) Electrons	d) Photons	e) None of these
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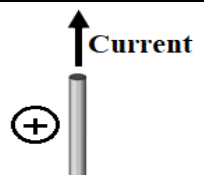
12- If $x(t) = (4 \text{ m}) \cos(3\pi t + 5\pi)$. The angular velocity is

a) 3π	b) 5π	c) $5\pi t$	d) 3π	e) None of these
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13- Which of the following represents a magnetic force:

a) $q \vec{v} \times \vec{E}$	b) $I \vec{B} \times \vec{L}$	c) $q B v \sin \theta$	d) $q \vec{B} \times \vec{v}$	e) None of these
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14- A magnetic force direction on a charge moving up as in Figure is:



a) + z	b) - x	c) - z	d) + y	e) None of these
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15- Two charges on x axis: q_1 at $x = a$ and q_2 at $x = - 2a$. The net force on a third charge placed at the $x = 0$ is zero, q_1 and q_2 must be related by:

a) $q_2 = q_1/4$	b) $q_2 = - q_1/2$	c) $q_2 = 4 q_1$	d) $q_2 = - 2q_1$	e) None of these
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Section C: Solve the following problems and select the proper choice:**20 Marks****Note: Show the full steps clearly for problems 19 and 20****16-** 9 electrons and 3 protons have a total charge of:

a) $11.2 \times 10^{-19} \text{ C}$	b) $-9.6 \times 10^{-19} \text{ C}$	c) $9.6 \times 10^{-19} \text{ C}$	d) $-11.2 \times 10^{-19} \text{ C}$	e) None of these
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17- In SHM: If a period is 2 s. The frequency is:

a) 0.05 Hz	b) 5 Hz	c) 50 Hz	d) 0.5 Hz	e) None of these
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18- A wire has current of 24 A and 0.75 m along the -y within a magnetic field $\vec{B} = 1.6 \hat{k} \text{ (T)}$.

The magnetic force on the wire is:

a) $-28.8 \hat{i} \text{ (N)}$	b) $-28.8 \hat{j} \text{ (N)}$	c) $28.8 \hat{i} \text{ (N)}$	d) $28.8 \hat{j} \text{ (N)}$	e) None of these
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19- A block whose mass $m = 500 \text{ g}$ is fastened to a spring with k is 3 N/m. The angular velocity is:

a) 8 rad/s	b) 3 rad/s	c) 2 rad/s	d) 4 rad/s	e) None of these
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20- A charge of - 4 C moves with velocity of 20 m/s directed to -z axis in magnetic field of 3 T directed to x axis. The magnitude and direction of magnetic force is:

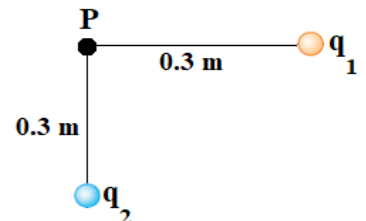
a) 24 N at y	b) -240 N at y	c) -24 N at -y	d) 240 N at y	e) None of these
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21- A proton moving with a velocity of $\vec{v} = 4 \hat{i} + 4 \hat{j} + 2 \hat{k} \text{ (m/s)}$ in $\vec{B} = 2 \hat{i} + 4 \hat{j} - 6 \hat{k} \text{ (T)}$. The magnitude of the magnetic force is:

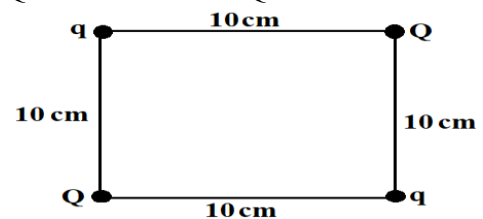
a) $31.9 \times 10^{-19} \text{ N}$	b) $71.6 \times 10^{-19} \text{ N}$	c) $69.2 \times 10^{-19} \text{ N}$	d) $100.2 \times 10^{-19} \text{ N}$	e) None of these
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22- A wire with mass per unit length of 0.04 kg/cm carries a current of 2 A moving to the right placed in magnetic field of 2 T directed outward a paper. The tension in the rope required to make the wire stable above the ground is 8.8 N. The length of wire will be:

a) 2.2 m	b) 2 m	c) 20 m	d) 0.2 m	e) None of these
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23- Given that $q_1 = q_2 = 4 \text{ nC}$ in the figure. The magnitude of the total electric field at P is:

a) 565.7 N/C	b) 707.7 N/C	c) 320000 N/C	d) 424.3 N/C	e) None of these
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24- A square is shown. If the total electric force on a charge Q is zero. Then Q is:

a) $6.33/q$	b) $-5.66 q/2$	c) $2.83 q$	d) $-q/4.33$	e) None of these
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25- 2 m cube each side is placed in $\vec{E} = 2 \hat{i} + 4 \hat{j} + 3 \hat{k} \text{ (N/C)}$. The electric flux in xy plane is

a) $12 \text{ N.m}^2/\text{C}$	b) $6 \text{ N.m}^2/\text{C}$	c) $-12 \text{ N.m}^2/\text{C}$	d) $-6 \text{ N.m}^2/\text{C}$	e) None of these
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Odd semester Examination – Academic year 2023/2024

Subject Physics II

Class B₁ITE-B₁EE

Date 11/ 06 /2024

Max. Marks 50

Time 2 hours

Lecturer Dr. Mehdi Dabban

Signature

Student Name

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Class

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Reg. No

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$$\text{Proton: } q = 1.6 \times 10^{-19} \text{ C, } m = 1.67 \times 10^{-27} \text{ kg}$$

$$\text{Electron: } q = -1.6 \times 10^{-19} \text{ C, } m = 9.11 \times 10^{-31} \text{ kg}$$

Section A: Answer the following questions with **True (T)** or **False (F)**:

14 Marks

1- A current flows out of a paper, a direction of B is counter clockwise	(T)	()
2- If a mass hanged to spring increases, the period decreases	()	(F)
3- If a charge enters parallel to B, the magnetic field is zero	()	(F)
4- The electron has positive charge	()	(F)
5- The magnitude of the velocity can be changed by electric force	(T)	()
6- The magnetic field is maximum at the poles	(T)	()
7- The electric flux through closed surface is zero	()	(F)

Section B: Select the proper choice for each of the following:

16 Marks

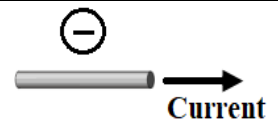
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a) 7π	b) $7\pi t$	c) 4π	d) $4\pi t$	e) None of these
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a) + y	b) + x	c) + z	d) - y	e) None of these
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11- To make a neutral object having positive charge, we add

a) Neutrons	b) Protons	c) Electrons	d) Photons	e) None of these
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12- We represent the magnetic field by:

a) F	b) q	c) B	d) E	e) None of these
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13- In SHM: The period can be given as:

a) $2\pi/f$	b) $2\pi\sqrt{\frac{k}{m}}$	c) $2\pi/\sqrt{\frac{m}{k}}$	d) $2\pi\sqrt{\frac{m}{k}}$	e) None of these
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14- Which of the following represents a magnetic force:

a) $I \vec{v} \times \vec{B}$	b) $q \vec{v} \times \vec{E}$	c) $I B v \sin \theta$	d) $I B L \sin \theta$	e) None of these
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15- Two charges on x axis: q_1 at $x = a$ and q_2 at $x = -2a$. The net force on a third charge placed at the $x = 0$ is zero, q_1 and q_2 must be related by:

a) $q_1 = 4 q_2$	b) $q_1 = -2 q_2$	c) $q_2 = q_1/2$	d) $q_1 = q_2/4$	e) None of these
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Section C: Solve the following problems and select the proper choice:

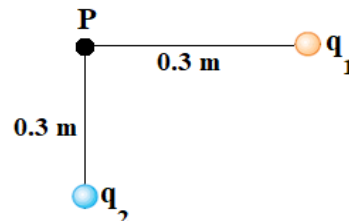
20 Marks

Note: Show the full steps clearly for problems 19 and 20

16- 2 m cube each side is placed in $\vec{E} = 2\hat{i} + 4\hat{j} - 3\hat{k}$ (N/C). The electric flux in xy plane is

a) 12 N.m ² /C	b) 6 N.m ² /C	c) - 12 N.m ² /C	d) - 6 N.m ² /C	e) None of these
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17- Given that $q_1 = q_2 = 3$ nC in the figure. The magnitude of the total electric field at P is:



a) 180000 N/C	b) 707.7 N/C	c) 424.3 N/C	d) 282.8 N/C	e) None of these
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18- A block whose mass $m = 500$ g is fastened to a spring with k is 8 N/m. The angular velocity is:

a) 8 rad/s	b) 3 rad/s	c) 2 rad/s	d) 4 rad/s	e) None of these
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19- A wire has current of 24 A and 0.75 m along the z within a magnetic field $\vec{B} = 1.6\hat{k}$ (T). The magnetic force on the wire is:

a) - 28.8 $\hat{i}$ (N)	b) - 28.8 $\hat{j}$ (N)	c) 28.8 $\hat{i}$ (N)	d) 28.8 $\hat{j}$ (N)	e) None of these
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20- 2 electrons and 9 protons have a total charge of:

a) 11.2×10^{-19} C	b) -9.6×10^{-19} C	c) 9.6×10^{-19} C	d) -11.2×10^{-19} C	e) None of these
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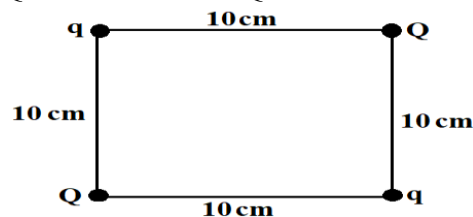
21- In SHM: If a period is 0.2 s. The frequency is:

a) 0.05 Hz	b) 5 Hz	c) 50 Hz	d) 0.5 Hz	e) None of these
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22- A wire with mass per unit length of 0.04 kg/cm carries a current of 2 A moving to the right placed in magnetic field of 2 T directed outward a paper. The tension in the rope required to make the wire stable above the ground is 13.2 N. The length of wire will be:

a) 30 m	b) 3 m	c) 0.3 m	d) 3.33 m	e) None of these
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23- A square is shown. If the total electric force on a charge Q is zero. Then Q is:



a) - q/4.33	b) 2.83 q6	c).33/q	d) - 5.66 q/2	e) None of these
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24- A proton moving with a velocity of $\vec{v} = 4\hat{i} + 2\hat{j} + 3\hat{k}$ (m/s) , in $\vec{B} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ (T). The magnitude of the magnetic force is:

a) 31.9×10^{-19} N	b) 71.6×10^{-19} N	c) 69.2×10^{-19} N	d) 100.2×10^{-19} N	e) None of these
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25- A charge of - 4 C moves with velocity of 2 m/s directed to -z axis in magnetic field of 3 T directed to x axis. The magnitude and direction of magnetic force is:

a) 24 N at y	b) -240 N at y	c) -24 N at -y	d) 240 N at -y	e) None of these
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